Two-player handheld: reference design (REF-A)

RP2354B, 3.5in 320x480 IPS (ST7796, SPI), 16 membrane contacts under DS Lite silicone,
PWM audio into an FM8002A, one Li-ion cell on a TP4056 charger with load sharing, ME6211 3V3.

Every sheet is generated from board/design.py by tools/build_sch.py. Edit design.py, not the sheets.

USB_Charger

usb_charger.kicad_sch

USB-C sink, ESD, TP4056 charger, load sharing, battery connector

Power

power.kicad_sch

ME6211 3V3 LDO, power button latch, battery and button sensing

MCU

mcu.kicad_sch

RP2354B (2 MB flash inside), 12 MHz crystal, USB series resistors

MCU_Power

mcu_power.kicad_sch

RP2354B decoupling and its internal 1.1 V switcher

Debug

debug.kicad_sch

SWD and UART connectors, BOOTSEL and RESET buttons

Display

display.kicad_sch

48-pin solder FPC of the 3.5in IPS panel, AP3156 backlight driver

Audio

audio.kicad_sch

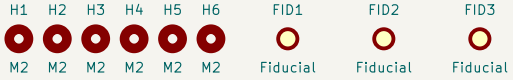
PWM low-pass into an FM8002A bridge amplifier, speaker pads

Buttons

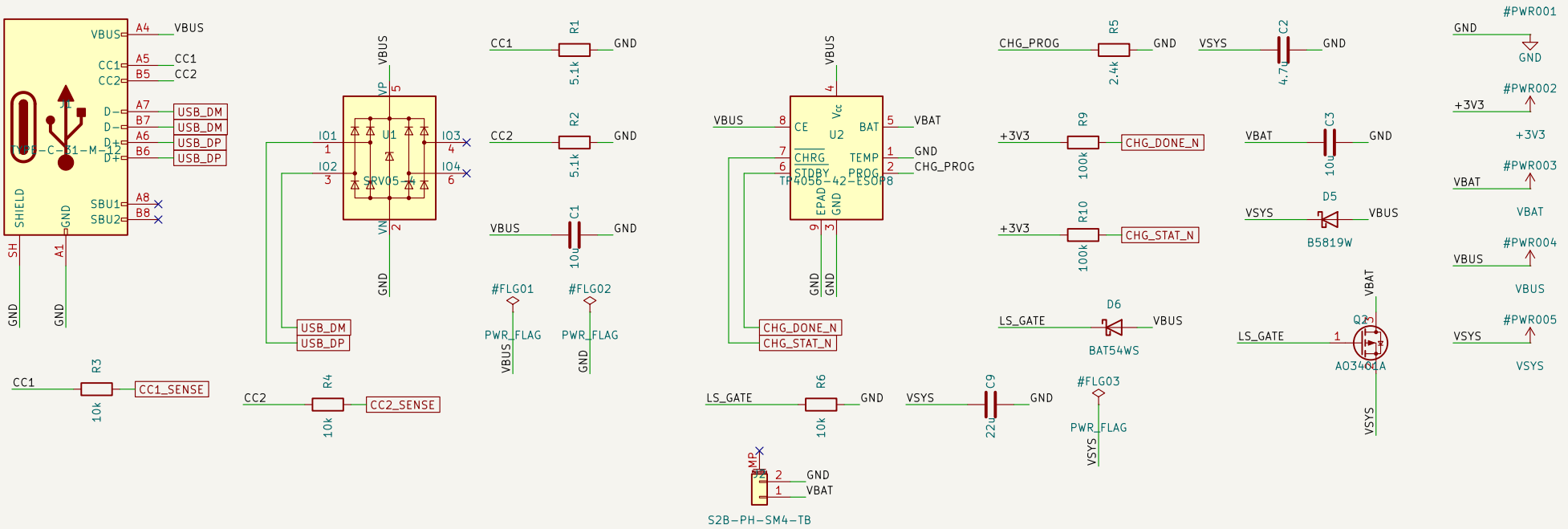
buttons.kicad_sch

16 membrane contacts, one GPIO each

Mechanical: locating holes at positions set by the case; fiducials for assembly.

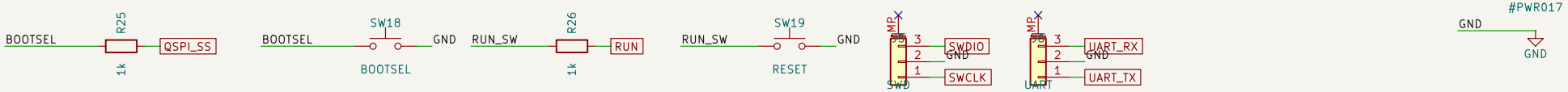


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TP4056 charges at 500 mA (R5); TEMP is grounded (no thermistor). USB feeds the system through D5;
without USB the cell does, through Q2. D6 keeps Q2's gate off C1, so R6 turns Q2 on at once on an unplug.
CC1/CC2_SENSE let firmware read the source's current offer (dim the backlight on a 500 mA port).

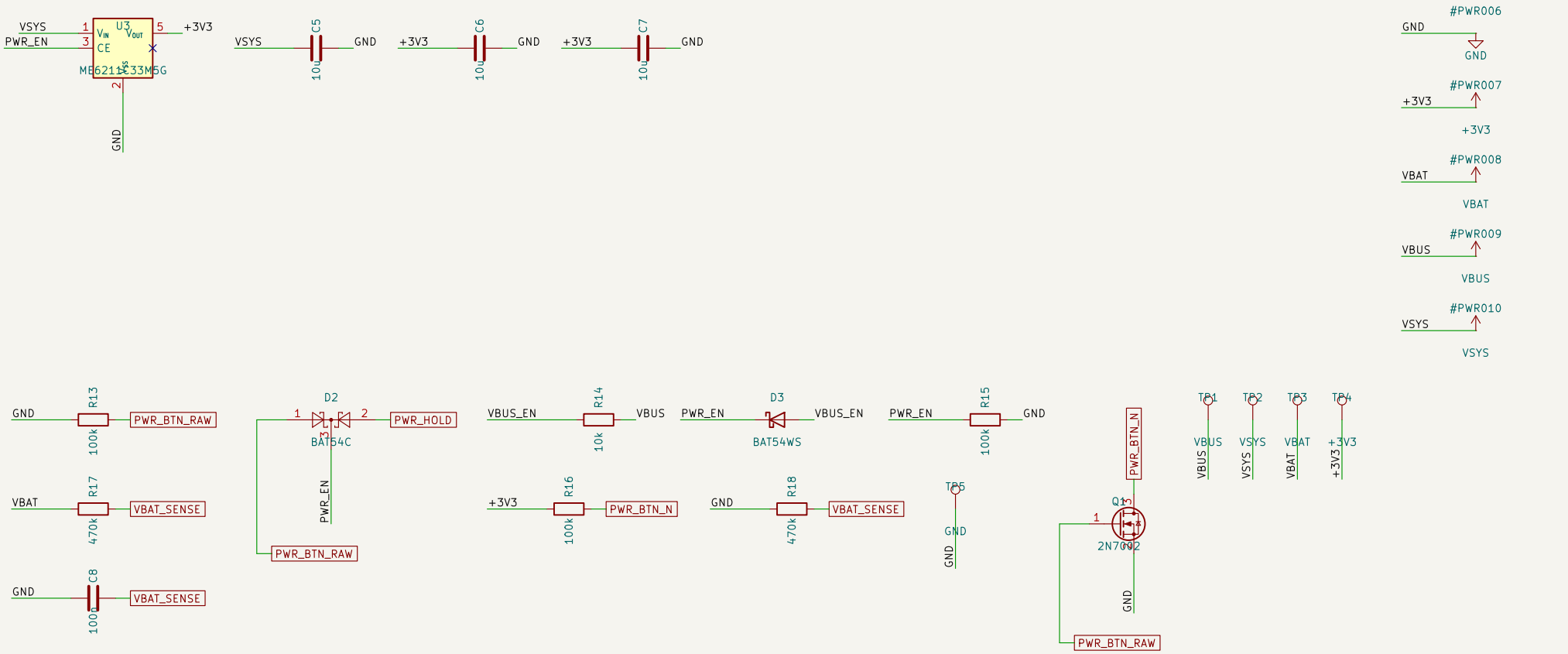
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File: usb_charger.kicad_sch		
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J5 SWD and J6 UART follow the Raspberry Pi 3-pin debug connector pinout (Debug Probe cable).

SW18 BOOTSEL and SW19 RESET sit under pinholes in the lid.

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Power: the button (seat 1's Y contact), the MCU (PWR_HOLD) or USB raise PWR_EN. Releasing it with PWR_HOLD low turns the device off. A reset drops PWR_HOLD, so a crash turns it off.

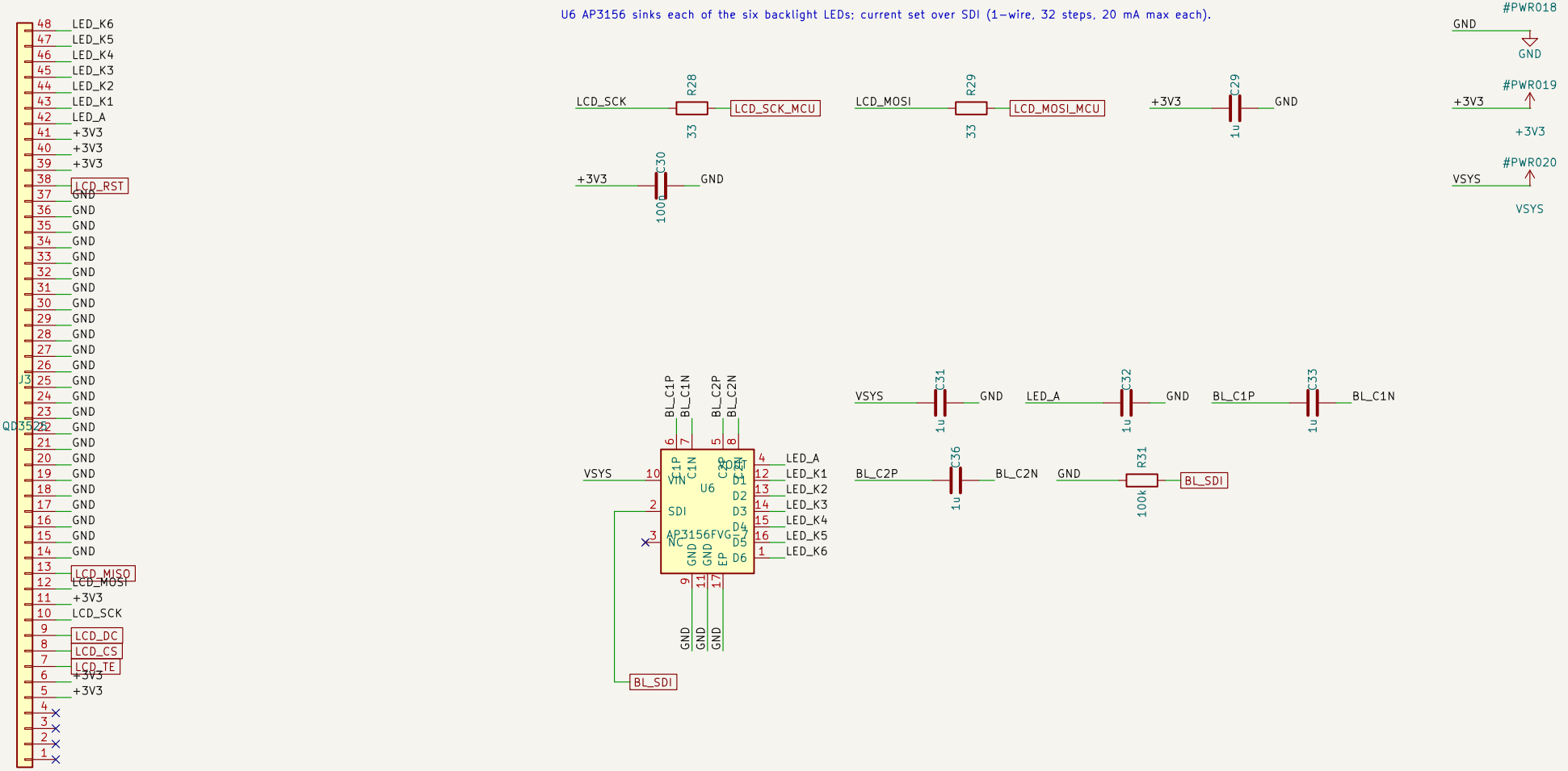
Q1 lets the MCU see the button (PWR_BTN_N low = pressed) without exposing a pin to VSYS.

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QD3525 3.5in IPS panel (ST7796U, 4-line SPI). Its 48-pin FPC folds under the panel and is soldered to J3.

Unused bus pins are tied as the panel maker's own module ties them: DB0-17 and RGB low, RD and IM2-0 high.

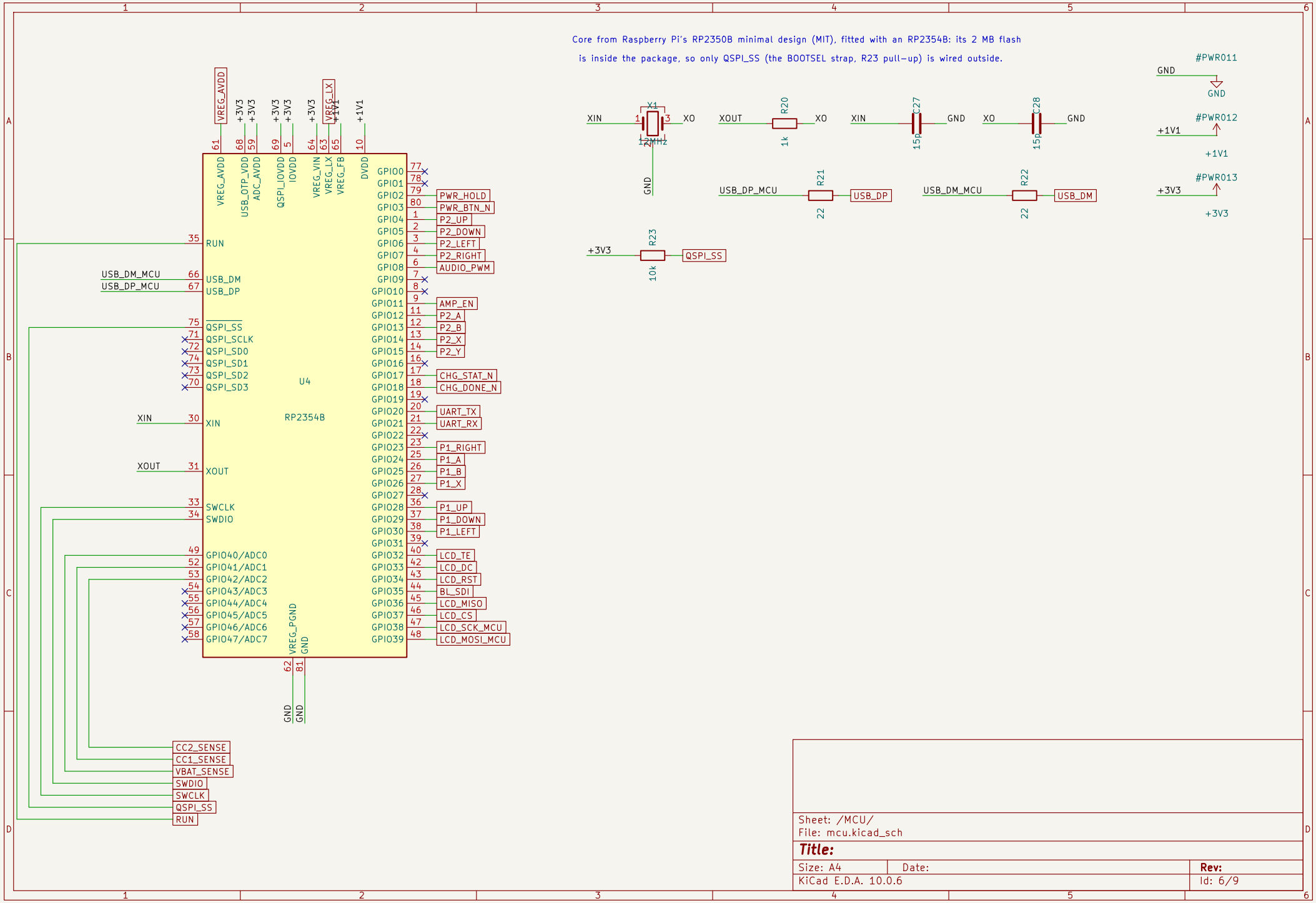
U6 AP3156 sinks each of the six backlight LEDs; current set over SDI (1-wire, 32 steps, 20 mA max each).

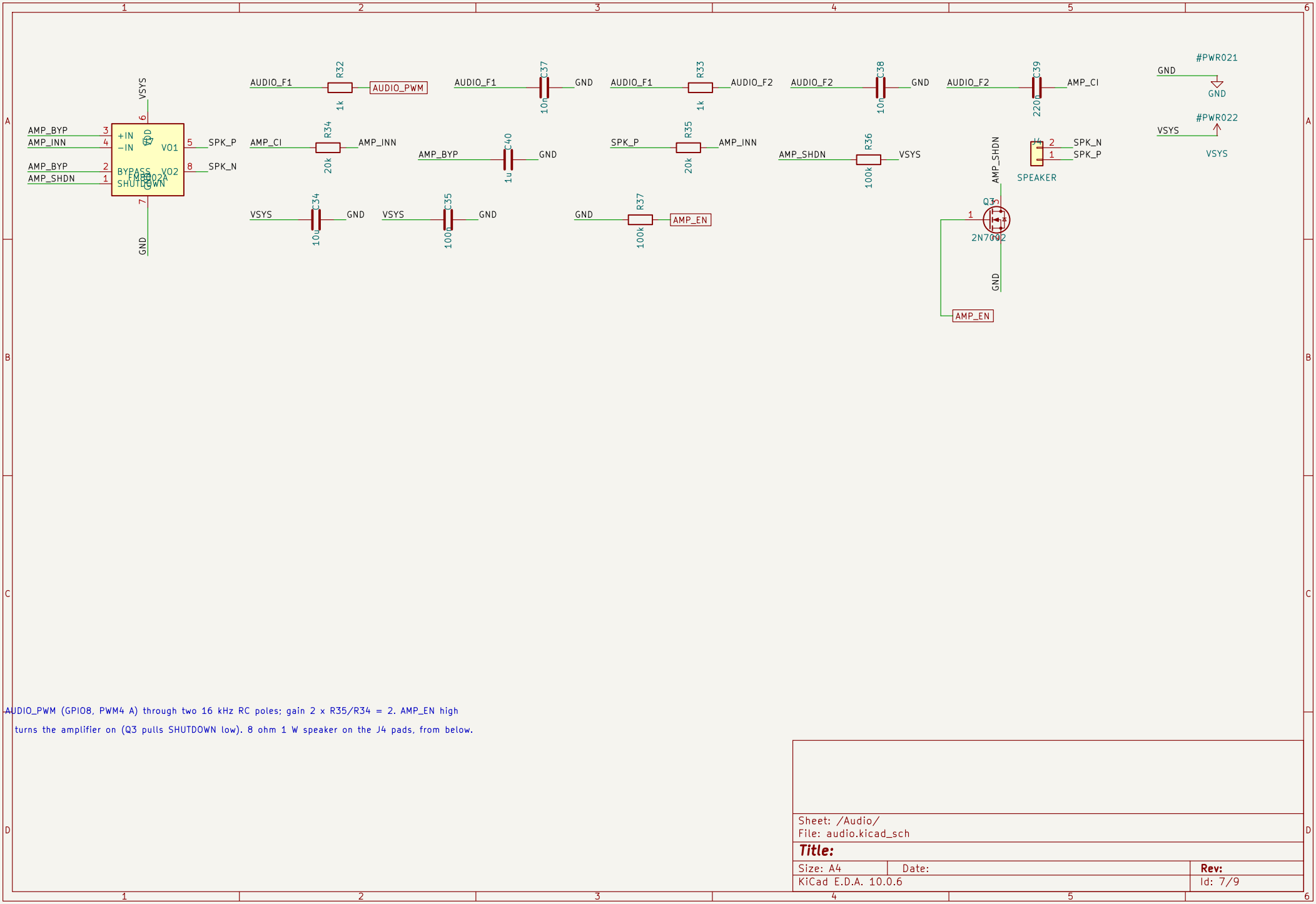


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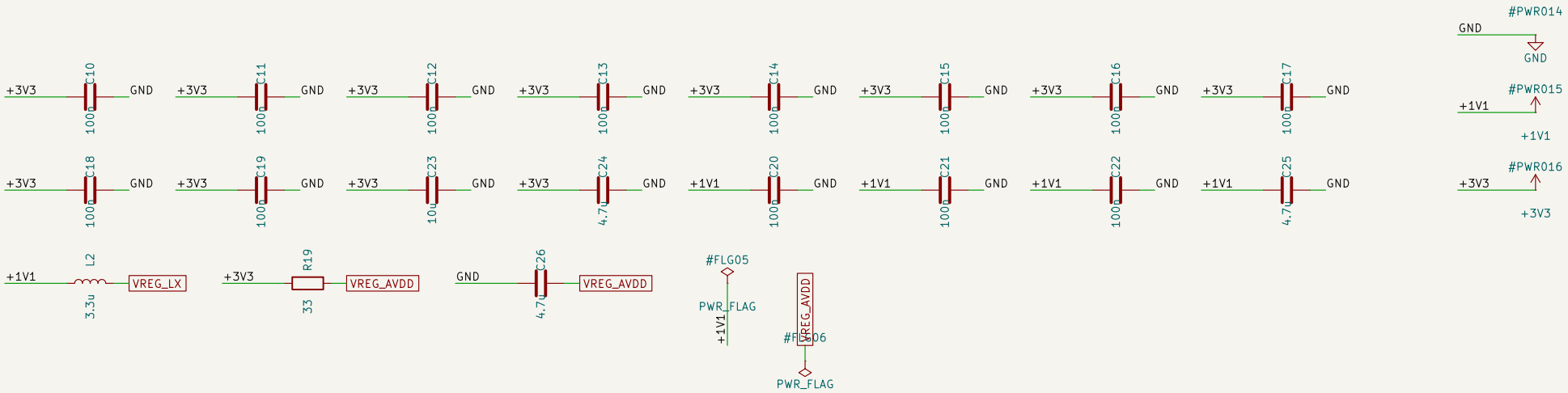
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Core from Raspberry Pi's RP2350B minimal design (MIT), fitted with an RP2354B: its 2 MB flash
is inside the package, so only QSPI_SS (the BOOTSEL strap, R23 pull-up) is wired outside.
U4
RP2354B
X1
R20
1k
C27
15p
C28
15p
R21
22
R22
22
R23
10k
#PWR011
#PWR012
#PWR013
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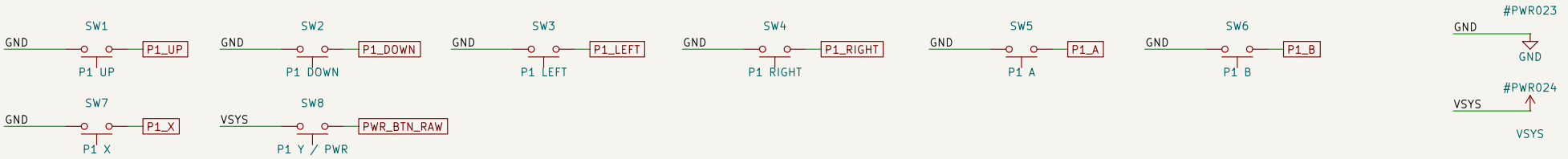
AUDIO_PWM (GPIO8, PWM4 A) through two 16 kHz RC poles; gain 2 x R35/R34 = 2. AMP_EN high turns the amplifier on (Q3 pulls SHUTDOWN low). 8 ohm 1 W speaker on the J4 pads, from below.

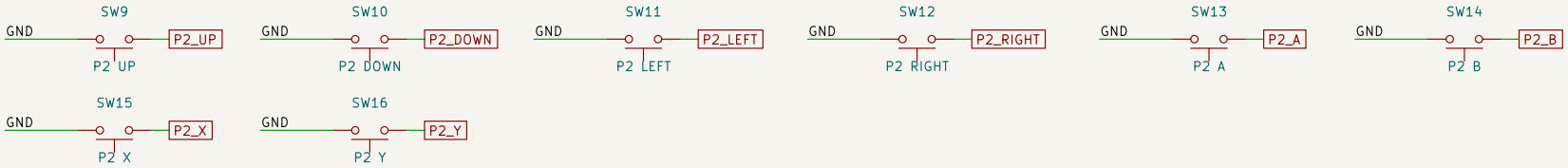
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RP2354B supply, as in Raspberry Pi's minimal design: 100 nF at every IOVDD, DVDD, ADC, OTP and QSPI pin, 4.7 uF at the regulator's input and output, and L2 (a polarity-marked 3.3 uH inductor, fit as drawn).

Sheet: /MCU_Power/	
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Bare ENIG contacts (DSL_button_4ring) under DS Lite silicone; the firmware enables pull-ups.

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